



Identifying Development Frameworks



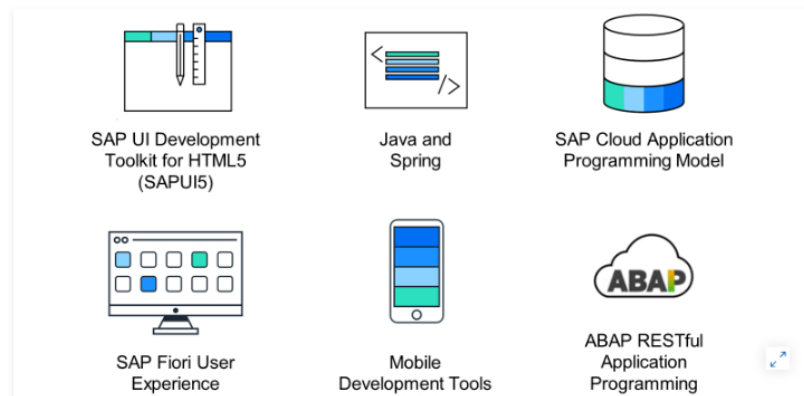
Objectives

After completing this lesson, you will be able to:

- Analyze development frameworks.
- Introduce SAP BTP Guidance Framework.

Development Frameworks

Example of Development Frameworks from SAP



SAP's strategy is to fully unify this developer experience for the business expert, citizen developer, and professional developer.

Today, SAP has several developer tools and frameworks supporting the full stack for UX, Data services, workflows programming models, and much more.

Here is a small extract from the developer tools:

SAP UI Development Toolkit for HTML5 (SAPUI5)

SAP UI Development Toolkit for HTML5 (SAPUI5) is based on the model view controller principle. SAPUI5 is a modern and flexible framework to build web applications for your business needs. In addition, there are tools for planning SAPUI5 apps, testing, icons, and more.

Learn more about [SAPUI5](#).

You can also use OpenUI5, the open source-based UI5 version.

Learn more about [OpenUI5](#).

Java and Spring

Experience

Development Tools

Programming

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Java and Spring

Through the Cloud Foundry or Kyma runtime, you are able to develop Java-based software development projects and operate them directly on SAP BTP. This has the positive effect that your Java project can get integrated easily into your enterprise lands. SAP Cloud Application Programming Model and your application can consume SAP BTP services. In addition, you can also use frameworks like Spring or SAP Cloud Application Programming Model.

Learn more about [Spring](#).

SAP Fiori User Experience

The SAP Fiori design system includes design languages for dedicated technologies: web, native mobile, and conversational. Each design language specifies the look and feel, controls and floor-plans and common functions, and includes guidelines and stencils for designers and developers.

SAP Fiori can be built using many technologies, with SAPUI5 being our reference HTML5 web technology. We have two native mobile design languages: one for iOS and one for Android.

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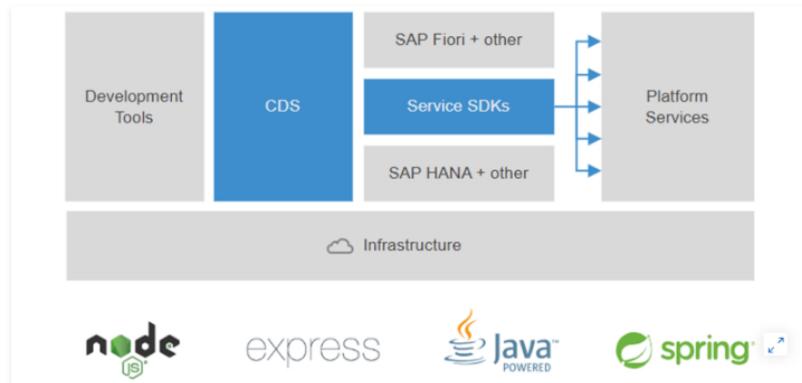
Learn more about [User Experience](#) and [SAP Fiori in our Learning Journey](#).

More information about [SAP Fiori](#).

SAP's Mobile Development tools

When you want to create mobile apps, today there are a bunch of technologies that can do that. SAP offerings:

versatile option for various project needs. You can combine a SAP Cloud Application Programming Model project with SAP Fiori projects to create user-friendly interfaces. Additionally, SAP Cloud Application Programming Model supports both Node.js and Java stacks, allowing developers to use the technology best suited to their needs. It facilitates the creation of OData services, which are essential for integrating different systems and providing a standardized approach to data access and manipulation.



The main components of SAP Cloud Application Programming Model include:

- **Core Data Services (CDS):** A universal modeling language for defining data models and services on a semantic level.
- **SAP Cloud Application Programming Model Services:** Supports the creation of services based on these models, providing mechanisms for data access, business logic, and service orchestration.
- **Event-Handlers:** Enable the implementation of custom business logic beyond standard CRUD operations.

A typical development workflow in SAP Cloud Application Programming Model include:

- **Project Initialization:** Use SAP Cloud Application Programming Model CLI tools to create and configure a new project.
- **Modeling:** Define data models and services using CDS.
- **Business Logic Implementation:** Write handlers and custom logic in Node.js or Java.
- **Deployment:** Deploy the application on SAP Business Technology Platform in the Cloud Foundry environment or other supported environments.

Cloud Application Programming Model projects integrate easily with various SAP technologies, including SAP Fiori for building user-friendly interfaces, SAP BTP for platform services, and SAP HANA for database services. Developers can use annotations in CDS models to automate the generation of SAP Fiori UIs, simplifying the creation of interactive and consistent user experiences based on OData services. Importantly, the OData services created using SAP Cloud Application Programming Model can be consumed by many other frontend or user interface technology beyond SAP Fiori, including SAP Build products and third-party UI frameworks like React, Angular, and Vue.js.

Learn more about [SAP Cloud Application Programming Model](#).

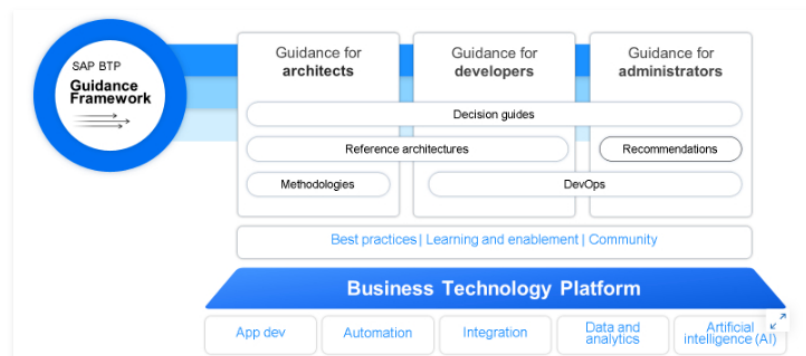
Learn more about [SAP HANA CDS](#).

Learn More

Learn more [about ABAP CDS](#).

You can find a lot of [SAP tools for software development purpose here](#) or [more details here](#).

SAP BTP Guidance Framework



SAP BTP Guidance Framework is your compass for architecting, implementing, and operating solutions on SAP BTP. It provides a central access point for decision guides, reference architectures, methodologies, and many best practices. The framework consists of the following elements:

- Decision guides helping you to identify the technology option that best fits your needs
- Reference architectures that allow you to explore how to architect solutions on SAP BTP with the help of curated solution templates
- Methodologies that offer a guided approach toward increasing your cloud technology maturity level, covering both technical and organizational aspects
- Recommendations for different technology domains, helping you to evaluate and choose the right implementation option
- DevOps principles that enable your development and operations teams to be more efficient

Get started with exploring best practices, grow your skills with learning and enablement offerings for SAP BTP, and engage with us in the SAP Community. Make sure to bookmark the SAP BTP

Guidance Framework and keep an eye on it, as it gets updated and enhanced from time to time.

You can access the SAP BTP Guidance Framework here:

[SAP BTP Guidance Framework](#) 

Key Takeaways Of This Lesson

SAP has a lot of developer tools and frameworks which provide the full development and user experience. SAP BTP Cockpit is the central entry point for managing your SAP BTP accounts. In addition, there are the ABAP Development Tools for developers who program ABAP in the IDE Eclipse and SAP Business Application Studio (BAS), which is a powerful and modern development environment in the cloud tailored for efficient development of business applications.

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